2024 Midterm: corrected worked answers

EE3032 Introduction to VLSI, National Central University

The two photographed prompt pages and all four supplied handwritten-answer pages were read before this report. The source sheet is graded 100, but its grade is not treated as a correctness certificate. Correct reasoning and diagrams are retained. The main numerical correction is Q6's gate-capacitance unit; Q3 also needs an explicit assumption about NAND input capacitance. All computations here use the prompt's ideal models, rather than an invented process simulation.

1. Brief answers (30 points)

(a) **Why not simply increase V_{DD}?** Supply voltage is limited by the technology's rated operating and reliability limits: stronger oxide electric field can cause breakdown and accelerate wear, and higher voltage increases switching energy roughly as CV_{DD}^2. Raising voltage is therefore not a free speed improvement. Retain the student's $E_{ox} \simeq V/t_{ox}$ reliability argument and add the power cost.

(b) **Mask-pattern transfer.** Clean/prepare the surface; coat and bake photoresist; align the mask and expose; develop the resist; transfer the pattern to the underlying film or silicon by etching; strip the resist. If the patterned opening is for implantation, the patterned resist instead serves as its mask. The student's coat/expose/etch sequence is useful, but needs development and resist removal.

(c) **FinFET sizing.** Increase or decrease the number of parallel fins (and permitted fingers/multiplicity). At fixed technology fin geometry, a tri-gate device has approximately $W_{eff} = N_{fin}(2H_{fin} + T_{fin})$; width is quantized. Retain the student's fin-count choice rather than freely changing a planar width. Fin geometry and permitted lengths are technology constrained. See the Synopsys reference below.

(d) **Two static-CMOS features.** It is non-ratioed for Boolean correctness, and provides full rail-to-rail restored output levels. Device ratios still affect delay and noise margins. Retain both listed features.

(e) **Two static-current mechanisms.** Subthreshold source–drain leakage in nominally off devices, and reverse-biased source/drain–body junction leakage. These are the two requested course mechanisms. Gate leakage can additionally matter in some technologies, but does not replace the requested pair.

(f) **Chip-on-wafer on a silicon interposer.** Prepare the interposer wafer with TSVs and redistribution/interconnect layers; provide matching fine-pitch die/interposer bump structures; prepare and singulate the known-good dies; align and bond each die face down to the interposer wafer; underfill/protect and test the assembly. Backside thinning/TSV reveal and backside contacts complete the interposer's external connection, with detailed ordering depending on the process. The later wafer-on-substrate assembly is a separate CoWoS step. The student's TSV, microbump and face-to-face assembly ideas are retained; no unsupported proprietary fixed sequence is claimed.

(g) **Backside power delivery.** Route supply/ground distribution on the wafer backside and connect it to the device power terminals using backside vias/contacts. Two advantages are reduced supply voltage drop from a less congested power path, and more frontside routing capacity for signals. Retain the student's separation of power and signal routing. Intel's PowerVia reference supports these benefits.

(h) **Body effect near the output.** In a conducting stacked pull-down, bulk is tied to VSS but the source of the upper transistor can rise above VSS, so V_{SB} and V_T increase. The output-side device can have the largest source–body reverse bias. A stacked pMOS pull-up has the analogous well/source effect. The explanation depends on internal source voltage and body connection, not merely on naming a device's saturation region. The statement applies to the usual biased stack; it is not a universal ranking at every transient instant.

(i) **Speed parameters.** From $t \sim C_L \Delta V / I_{drive}$, the principal circuit parameters are load capacitance, required voltage swing/supply, and available drive current. Current itself depends on W/L, mobility, oxide capacitance and overdrive/threshold. Retain the student's charge/current argument.

(j) **Output capacitance.** Sum the connected downstream gate input capacitances, output-node source/drain diffusion capacitances, and routing/interconnect capacitance: $C_L = C_{gate} + C_{diff} + C_{wire}$. Retain the student's three components.

2. MUX realization and unbuffered delay (15 points)

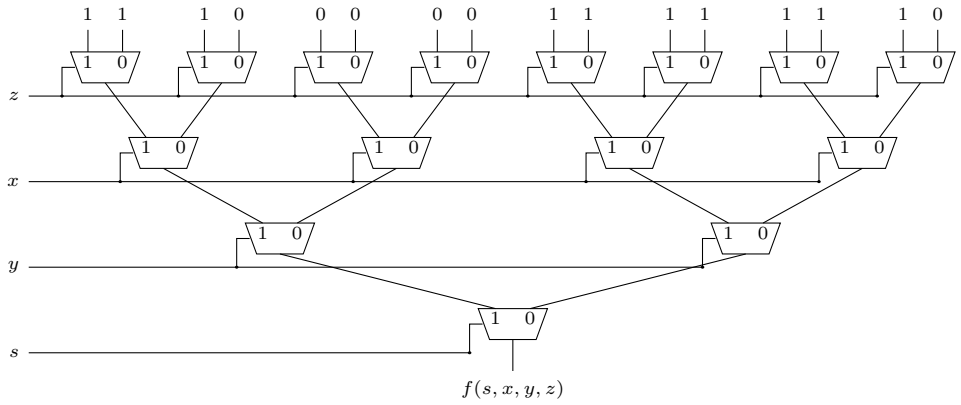
(a) Correct 16-to-1 assignment

$$f(s,x,y,z) = s\,y(x+z) + \bar{s}\,(x+y+z).$$

Retain the student's select order: leaf-to-root z, x, y, s. Each cell chooses its left input at select 1 and its right input at select 0. The sixteen data constants from left to right are

$$1,1,1,0,0,0,0,0, \quad 1,1,1,1,1,1,1,0.$$

The first eight implement $y(x+z)$ at $s=1$; the last eight implement $x+y+z$ at $s=0$. The tree has fifteen 2-to-1 cells and four stages. All sixteen assignments are independently checked by the published executable; the supplied leaf constants are correct.



(b) Delay without inserted buffers

The selected path consists of n series transmission-gate resistances. Nodes retain diffusion/wiring/load capacitance, so a signal must charge the downstream ladder through upstream resistance. Under an explicitly illustrative uniform ladder with each selected TG resistance R and each of its n output nodes capacitance C, its Elmore first moment is

$$\tau_E = \sum_{i=1}^{n} R \sum_{j=i}^{n} C = RC\frac{n(n+1)}{2} = \Theta(n^2).$$

For the four-stage path this model gives $10RC$. This is a first moment, not an exact 50% or 10–90% crossing time. Real coefficients depend on the node capacitances and nonlinear TG resistance; the quadratic scaling is the requested conclusion. The student correctly identifies a distributed RC ladder and quadratic stage dependence. In the ideal case of only a fixed final capacitor and zero internal capacitance, the delay instead scales as nRC_L; that is a different assumption.

3. Minimum-delay buffer chain (10 points)

Preserve polarity. A already inverts x, so the dashed inserted chain must be noninverting. Add an even number of inverters: the total number N of driving stages, including A, is odd.

The specified rise-time convention is $t_r = R_{p,D}C = 0.05RC$, so each NAND pMOS has resistance $R/20$ and width $20W/L$ for the worst single-on-pMOS rise. This alone does not specify the NAND nMOS widths or total per-input capacitance. Q3 defines C_g as the *entire input capacitance of the minimum inverter*, comprising its two W/L devices. If a NAND nMOS is w_nW/L, then

$$C_D = \frac{20 + w_n}{2} C_g, \qquad H = \frac{C_D}{C_g}.$$

This missing width is why $C_D = 20C_g$ cannot be uniquely inferred from the printed assumptions.

For equal-strength nMOS/pMOS inverter stages of size a_i relative to A, $a_0 = 1$, and negligible parasitics,

$$T = RC_g \left(\frac{a_1}{a_0} + \frac{a_2}{a_1} + \cdots + \frac{H}{a_{N-1}} \right).$$

For fixed N, equal fanout minimizes the sum: $h = H^{1/N}$, $a_i = h^i$, and $T_N = NH^{1/N}RC_g$. The continuous optimum is $N = \ln H$, with stage effort e; compare the allowed odd integers rather than inserting an arbitrary parity.

Explicit NAND assumption	H	$T_1/(RC_g)$	$T_3/(RC_g)$	$T_5/(RC_g)$
Equal NAND widths $w_n = 20$	20	20	8.1433	9.1028
Balanced 2-series nMOS $w_n = 40$	30	30	9.3217	9.8718

Both conventions select $N = 3$: A plus two inserted inverters. Under the **student's intended $H = 20$ assumption**, exact inserted widths for both nMOS and pMOS are $20^{1/3}W/L = 2.7144W/L$ and $20^{2/3}W/L = 7.3681W/L$. Using the supplied approximations gives the student's eW/L, e^2W/L and $T \simeq 3eRC_g$; retain this as an approximate conditional answer. For $H = 30$, the corresponding widths are $3.1072W/L$ and $9.6549W/L$.

There are three inverter stages on the path, but only two *inserted* inverters. This T ends at the input of NAND D, as drawn; D's additional output delay is fixed and does not alter the optimum under the same convention.

4. Feedback NMOS gate (10 points)

Before switching, $V_{out} = 1.2$ V turns on feedback device N2. In the stated constant-on-resistance approximation, after N1 turns on the internal source node u is

$$u = 1.2\frac{R}{R + 2R} = 0.4 \text{ V}.$$

N3 starts conducting when $V_{in} - u = V_{Tn} = 0.4$ V; therefore the input at x is $\boxed{V_{in}(t_2) = 0.8 \text{ V}}$. As output falls, N2's gate voltage falls, reducing its feedback pull-up and allowing the internal node to drop. This positive feedback assists the transition.

For the open interval $t_1 < t < t_2$, $0.4 < V_{in} < 0.8$ V and $u \simeq 0.4$ V:

Device	Bias check	Region
N1	$V_{DS} = 0.4 \geq V_{GS} - V_T = V_{in} - 0.4$	saturation
N2	$V_G = V_D = 1.2$, $V_S = 0.4$; $V_{DS} = 0.8 > 0.4$	saturation
N3	$V_{GS} = V_{in} - 0.4 < 0.4$	cutoff

At t_1 N1 is just at its threshold; at t_2 N3 reaches its threshold and N1 reaches its saturation/linear boundary in this approximation. Do not call N3 already conducting before t_2, or indefinitely cutoff after output falls. Retain the student's 0.8 V and interior-interval classification with these endpoint qualifications. The voltage divider uses the supplied effective-resistance model; it is not an exact MOS square-law DC solution with body effect.

5. Noise-margin comparison (10 points)

Every listed word makes $q = G1 = \overline{BC} = 1$ and $D = 1$:

$ABCD$	q	A	G2 output	G3 output
0011	1	0	0	0
0101	1	0	0	0
1101	1	1	0	0
1001	1	1	0	0

Because $W_n = W$, $W_p = 2W$ and $\mu_n = 2\mu_p$, the strength of one nMOS equals that of one pMOS: $\beta_n = \beta_p = \beta$ under common oxide/length and the course's matched-threshold comparison.

For **G2 (NOR)**, the worst active input transfer with the other input low has one nMOS pull-down versus two series pMOS pull-up devices: approximately $\beta_{n,eff} = \beta$, $\beta_{p,eff} = \beta/2$. The stronger pull-down moves the transition lower, giving $\boxed{NM_L < NM_H}$. If both inputs are high, two parallel nMOS only reinforce its low output. In the listed $A = 1, q = 1$ words, sweeping q alone while holding $A = 1$ leaves the output clamped low; it does not define a complete inverter VTC or a new pair of conventional noise margins. State the active/worst transfer convention explicitly.

For **G3 (NAND)**, the relevant single-input transfer with the other input high has two series nMOS versus one pMOS: $\beta_{n,eff} \simeq \beta/2$, $\beta_{p,eff} = \beta$. The stronger pull-up moves the transition higher, giving $\boxed{NM_L > NM_H}$. Retain both of the student's relationships under this effective-strength, matched-threshold model.

Formal margins are $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$; the VTC limits occur at slope -1. Strength ratios and nominal rail levels explain the direction, but the source supplies no thresholds/body parameters for unique numerical margins. The executable uses labeled illustrative square-law parameters only to check the direction, without presenting them as process measurements.

6. AND sizing below $8RC_g$ (10 points)

Correct the capacitance unit. Here C_g is the input capacitance of *one* W/L nMOS or pMOS, whereas in Q3 it was the whole minimum-inverter input capacitance. A W/L nMOS plus $2W/L$ pMOS inverter has $3C_g$, not $\frac{3}{2}C_g$.

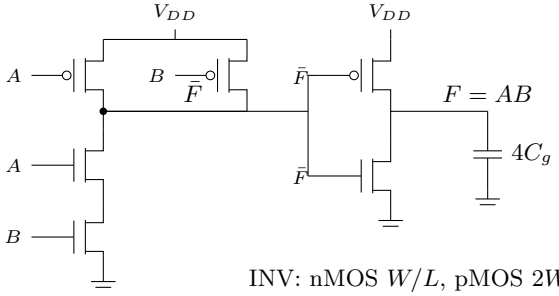
The student's drawn NAND uses W/L for each nMOS and $2W/L$ for each pMOS; its final inverter also uses nMOS W/L, pMOS $2W/L$. The worst NAND falling resistance is $2R$, so the complete AND rising delay is $2R(3C_g) + R(4C_g) = 10RC_g$, exceeding the requested bound. Its opposite delay is $R(3C_g) + R(4C_g) = 7RC_g$. Its topology is useful, but the claimed worst delay relies on the wrong capacitance unit.

Retain the inverter and each NAND pMOS at $2W/L$; increase **each NAND nMOS to** $2W/L$. Then each series nMOS has $R/2$, so the complete NAND pull-down resistance is R, matching the worst one-on-pMOS pull-up. Both changing-output directions have the bound

$$\boxed{T_{AND,worst} = R(3C_g) + R(4C_g) = 7RC_g < 8RC_g.}$$

Parallel-on pMOS cases can be faster. No diffusion or wire capacitance is counted because the prompt excludes it. The result uses the prompt's RC delay convention; it does not silently substitute a 0.69 factor.

NAND: both nMOS $2W/L$, both pMOS $2W/L$



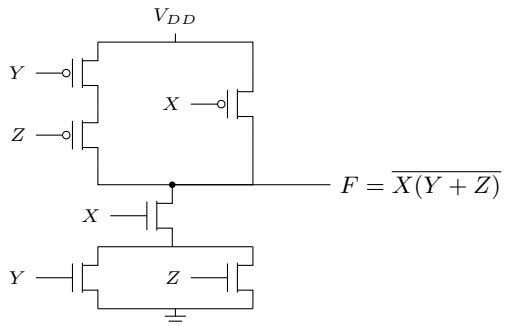
INV: nMOS W/L, pMOS $2W/L$

7. Static complementary CMOS (15 points)

Each full-expression bar is preserved from the printed prompt; there are no inner complemented input variables. The supplied handwritten networks have the correct series/parallel connectivity and are retained. All nMOS bodies are tied to VSS and all pMOS bodies to VDD. Each PDN realizes the expression under the bar; its dual PUN realizes the complement.

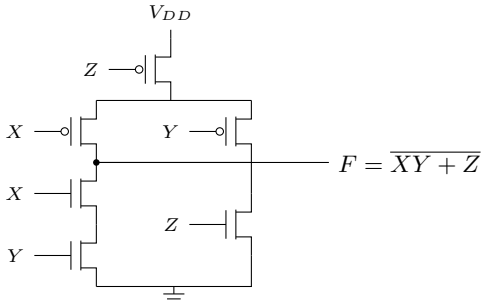
(a) $F = \overline{X(Y+Z)}$ – 6 transistors

PDN: X in series with Y parallel Z. PUN: X parallel the series pair Y,Z.



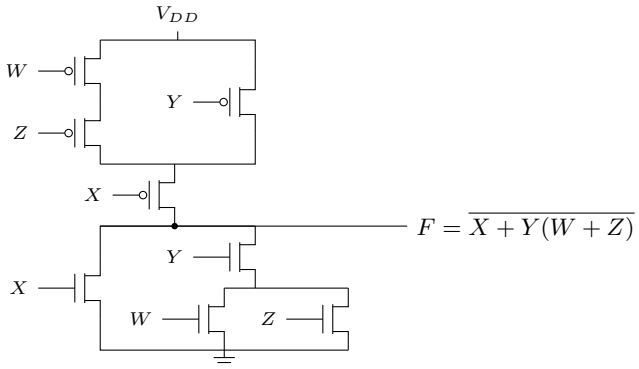
(b) $F = \overline{XY+Z}$ – 6 transistors

PDN: the series pair X,Y in parallel with Z. PUN: Z in series with X parallel Y.



(c) $F = \overline{X + Y(W + Z)}$ – 8 transistors

PDN: X in parallel with Y in series with W parallel Z. PUN: X in series with Y parallel the series pair W,Z.



The executable evaluates all 8, 8 and 16 input words using switch-graph connectivity. Exactly one of VDD or VSS is connected to F in every case, and its voltage polarity matches the original Boolean expression. All 32 checks pass; these are ideal connectivity checks, not a fabricated analog/PVT characterization.

Source reconciliation and reproduction

The student's correct Q2 constants, quadratic passive-delay explanation, Q4 nominal threshold/interior regions, Q5 strength comparison, and Q7 diagrams are retained. Q1 short answers are expanded where steps or physical conditions were omitted. Q3's graded $20C_g$ convention is explicitly conditional. Q6's capacitance mistake is corrected by strengthening the NAND nMOS devices, yielding $7RC_g$.

Run `python verify_q2_q7.py` and `python verify_q3_q6.py` in this folder. They regenerate the analytical CSV/JSON evidence. No process model is needed. Both this report and `exam.tex` are standalone pdfLaTeX documents; all circuits are native TikZ vectors, with no scan-image dependency. See `SOURCES.md`, `sources.json` and the original two-page extraction for provenance and limitations.

Primary technical references for Q1

1. Synopsys, FinFET Design Insights and Reliability: fin geometry, effective electrical width and quantized sizing. The general device facts are used, not its historical process-roadmap predictions.

2. Intel, PowerVia Test Shows Industry-Leading Performance, 2023: backside routing separates power from signal wiring and improves voltage droop. Product-release forecasts are not part of the answer.

3. TSMC, CoWoS technology: silicon-interposer die integration; detailed manufacturing order is technology dependent.

4. TSMC, US11094671B2, Package with thinned substrate, Figure 11: a disclosed example of die-to-interposer-wafer bonding, encapsulation, backside processing and later substrate assembly. This supports a representative sequence, not a universal commercial recipe.